DATA 255-12, Spring 2026

Assignment #5

Due by 3pm on March 8th, 2026

Notes

This assignment should be submitted in Canvas as a format of pdf generated by MS word or latex(overleaf) instead of handwriting (assignment4.pdf).

You may collaborate on homework but must **independently** write code/solutions. Copying and other forms of cheating will not be tolerated and will result in a **zero score** for the homework (minimal penalty) or a failing grade for the course. Your work will be graded in terms of correctness, completeness, and clarity, not just the answer. Thus, correct answers with no or poorly written supporting steps may receive very little credit.

1. (4 pts) Neural Network with one output neuron

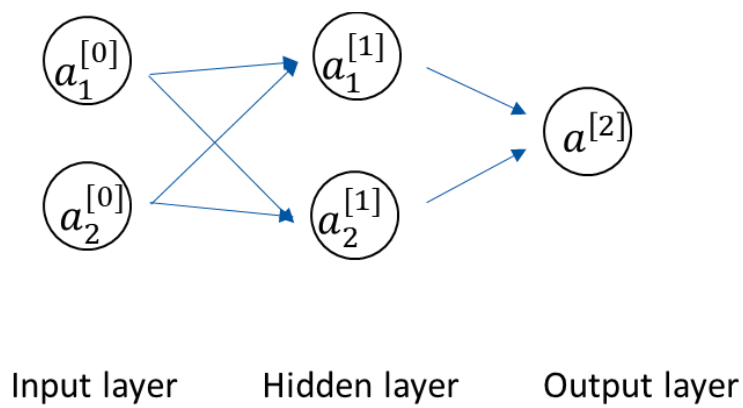

Fig.1 Architecture of a neural network with one output neuron

Please write backward propagation of the neural network.

In other words, please derive the following derivatives.

$\frac{\partial L(a^{[2]},y)}{\partial W_0^{[2]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_1^{[2]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_2^{[2]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_{10}^{[1]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_{11}^{[1]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_{12}^{[1]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_{20}^{[1]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_{21}^{[1]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_{22}^{[1]}}$,

NOTE:

The number of neurons in the input layer is 2.
The number of neurons in the hidden layer is 2.
The number of neurons in the output layer is 1.
Please assume that there is Bias term ($W_0^{[l]}$) for the hidden layer and the output layer.

For neurons in the hidden layer, please apply Sigmoid activation function.
For neurons in the output layer, please apply Sigmoid activation function.

Sigmoid activation function:

$$\sigma(z) = \frac{1}{1 + exp\ (-z)}$$

2. (6 pts) Neural Network with two output neurons

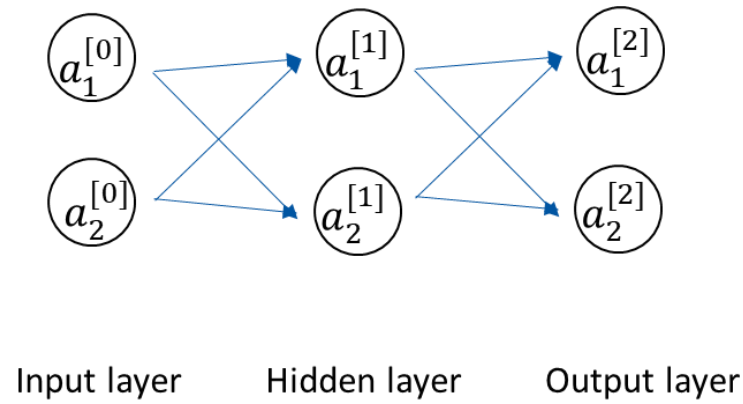

Fig.2 Architecture of a neural network with two output neurons

Please write backward propagation of the neural network.

In other words, please derive the following derivatives.

$\frac{\partial L(a^{[2]},y)}{\partial W_{10}^{[2]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_{11}^{[2]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_{12}^{[2]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_{20}^{[2]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_{21}^{[2]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_{22}^{[2]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_{10}^{[1]}}$,

$\frac{\partial L(a^{[2]},y)}{\partial W_{11}^{[1]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_{12}^{[1]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_{20}^{[1]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_{21}^{[1]}}$,
$\frac{\partial L(a^{[2]},y)}{\partial W_{22}^{[1]}}$,

NOTE:

The number of neurons in the input layer is 2.
The number of neurons in the hidden layer is 2.
The number of neurons in the output layer is **2**.
Please assume that there is Bias term ($W_0^{[l]}$) for the hidden layer and the output layer.

For neurons in the hidden layer, please apply Sigmoid activation function.
For neurons in the output layer, please apply **SoftMax** activation function.

SoftMax function:

$$\boldsymbol{a_1^{[2]}} = \frac{\boldsymbol{exp}\,(\boldsymbol{Z_1^{[2]}})}{\sum_{j=1}^{K} \boldsymbol{exp}\,(\boldsymbol{Z_j^{[2]}})}$$